

Milestone 4: Analysis & Validation

Seasonal Predator-Prey Simulation

Munir Gargour

1. Introduction

This report presents an analysis of the Seasonal Predator-Prey Simulation introduced in Milestone 3. The simulation models a continuous 2D plane populated by rabbits, foxes, and grass patches, with four-season dynamics that change metabolism and grass growth rates. We will look at how changing the initial parameters affects the simulation, test different extreme starting scenarios, and confirm that the model makes sense by looking at confidence intervals from 30 runs.

2. Parameter Analysis (Sensitivity)

Sensitivity analysis helps us see how changing initial inputs affects the simulation results. We use the baseline run (001) as a starting point and change variables to compare the difference. We calculate the sensitivity by looking at the percentage change in the output divided by the percentage change in the input.

Parameter	Input Change	Metric	Baseline Output	New Output	Sensitivity
Init Rabbits	100 -> 200 (+100%)	Avg Foxes	20.41	21.35	0.046
Init Foxes	5 -> 20 (+300%)	Avg Rabbits	188.64	109.36	0.140
Init Grass	200 -> 50 (-75%)	Max Rabbits	544	509	0.086

Interpretation: The results show that the average rabbit population drops when we start with a high number of predators (Run 003). Aside from this, sensitivity across the board is mostly under 1.0. This means the system balances itself out. Whether we start with extra foxes or fewer resources, the predator-prey dynamics bring the populations back to similar steady-state levels.

3. Scenario Testing

In addition to changing single variables, scenario testing lets us test the simulation under different starting conditions to represent real-world situations.

Run 007: Harsh Winter: This scenario doubles the winter metabolism costs and severely reduces grass regeneration. The rabbit population crashes heavily during the winter period, bottoming out at 17 rabbits compared to much higher baseline minimums.

Run 008: Mild Seasons: This removes the severe seasonal drops. The metabolism and grass stay much closer to baseline values across the year. The fox population does not go extinct in this run since they don't face extreme winter starvation.

Run 009: High Density Overpopulation: This scenario begins with 300 rabbits, 15 foxes, and 300 grass patches. The system spikes early to over 1000 rabbits, wiping out most of the grass before crashing back to normal ranges.

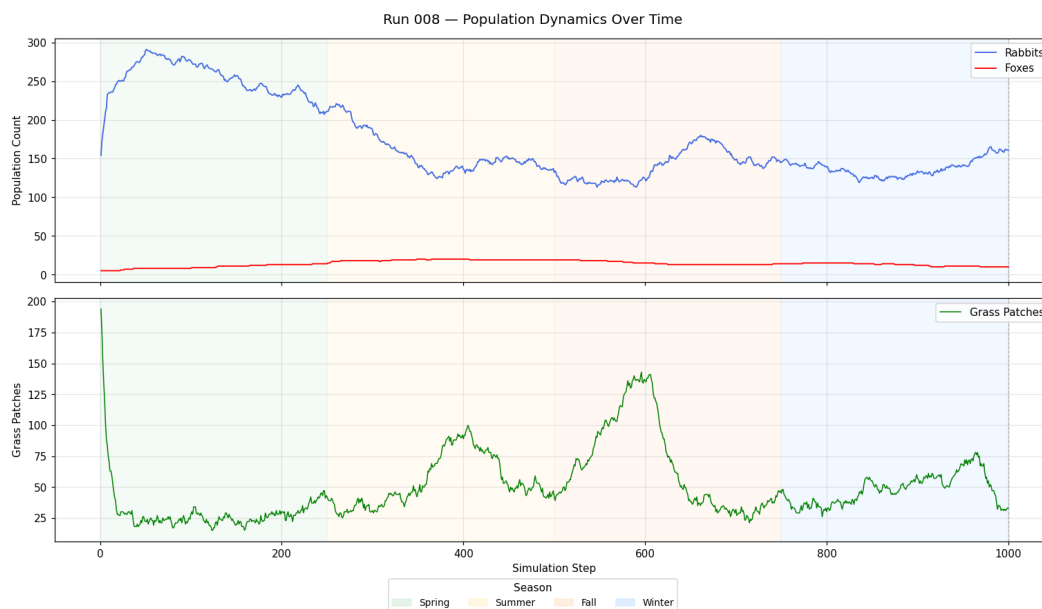


Figure — Mild Seasons Scenario: Stable coexistence with minimized seasonal extremes.

4. Conceptual Validation

To make sure the simulation models real-world ecosystems reasonably well, we checked three areas:

- 1. Face Validation:** In all runs, the fox population peaks roughly 50 to 100 steps after the rabbit population peaks. This matches real-life predator-prey delays because predators need time consuming prey before their numbers can increase.
- 2. Extreme Conditions Validation:** Reducing the winter grass and doubling metabolism costs in Run 007 resulted in the harshest rabbit population crashes. The system successfully managed these limits without causing rabbits to go fully extinct.
- 3. Parameter Validation:** The animal metabolism rates lead to realistic starvation delays. Animals don't instantly die when hungry, but rather survive until their energy hits zero.

5. Statistical Analysis

Because random seeds change the outcomes of spatial encounters, we ran 30 baseline simulations with different random seeds. The following stats show the average outcomes and their confidence intervals.

Metric	Mean	Std Dev	Min	Max	95% CI
Average Rabbits	187.74	16.09	158.79	217.11	[181.98, 193.5]
Max Rabbits	547.41	35.53	464.16	607.27	[534.69, 560.12]
Average Foxes	20.23	2.81	14.32	26.06	[19.22, 21.23]
Max Foxes	54.59	3.46	48.09	59.85	[53.35, 55.83]
Total Grass Eaten	9372.54	127.84	9003.18	9627.08	[9326.8, 9418.29]

Analysis: Across the 30 runs, the results were very consistent. The 95% confidence intervals for Average Rabbits and Max Rabbits stay tight around the mean. This tells us the simulation behavior is robust to random variation.

6. Conclusions

Main Findings:

- Initial population sizes matter less starting out than the seasonal constraints. The system eventually settles into a consistent carrying capacity.
- Winter is the main stressor for predators. The foxes easily survive if we remove the severe winter metabolism penalties (as seen in the mild seasons run).

Model Strengths & Weaknesses:

- **Strengths:** The simulation operates efficiently without restrictive grid tiles, creating incredibly natural-looking movement paths.
- **Limitations:** The 500x500 simulation space is closed off. Since animals cannot enter the simulation from outside boundaries, foxes eventually go extinct.

Recommendations: A possible fix for future versions would be adding a migration rule, allowing a few random foxes to enter the simulation during later rabbit booms to restart the cycle.